Series Solution of a Second Order ODE at $x_0 \neq 0$

- In general, $x_0 \neq 0$ can be *any* ordinary point of $y'' + P(x)y' + Q(x)y = 0$.
- The method is identical, but the series is expanded in powers of $(x - x_0)$ instead of x.
- Step 1: substitute $t = x - x_0$, so that $x = t + x_0$ and $t = 0$ corresponds to $x = x_0$.
- Step 2: rewrite $P(x)$ and $Q(x)$ as functions of t using $x = t + x_0$.
- Step 3: assume $y = \sum_{n=0}^{\infty} a_n t^n$ and substitute as before.
- Step 4: derive the recurrence relation in terms of a_n.
- Step 5: convert back to x using $t = x - x_0$ at the end.
- The radius of convergence is the distance from x_0 to the nearest singular point.



Problem 9: Setup ($x_0 = 1$)

- Solve $y'' + xy' + y = 0$ about the ordinary point $x_0 = 1$.
- Here $P(x) = x$ and $Q(x) = 1$ are both entire functions, analytic for every x.
- So $x_0 = 1$ is indeed an ordinary point (as is every other real number).
- Substitute $t = x - 1$, so $x = t + 1$; the equation becomes

$$y'' + (t+1)y' + y = 0$$

- Assume $y = \sum_{n=0}^{\infty} a_n t^n$.
- Then $y' = \sum_{n=0}^{\infty} n a_n t^{n-1}$, $\quad y'' = \sum_{n=0}^{\infty} n(n-1) a_n t^{n-2}$.
- We must expand $(t+1)y' = ty' + y'$ and match every power of t.
- This requires shifting three different series onto a common index k.



Problem 9: Deriving the Recurrence Relation

- Shift y'': let $n = k + 2$, giving $\sum_{k=0}^{\infty}(k+2)(k+1)a_{k+2}t^k$.
- Shift ty': let $n = k$ (no change needed), giving $\sum_{k=0}^{\infty} ka_k t^k$.
- Shift y': let $n = k + 1$, giving $\sum_{k=0}^{\infty}(k+1)a_{k+1}t^k$.
- The term y is already $\sum_{k=0}^{\infty} a_k t^k$.
- Adding all four series and setting the coefficient of t^k to zero:

$$(k+2)(k+1)a_{k+2} + ka_k + (k+1)a_{k+1} + a_k = 0$$

- Simplify: $ka_k + a_k = (k+1)a_k$, so

$$(k+2)(k+1)a_{k+2} + (k+1)a_{k+1} + (k+1)a_k = 0$$

- Divide by $(k+1)$: $\quad (k+2)a_{k+2} = -a_{k+1} - a_k$.



Problem 9: Final Solution

- Recurrence relation: $a_{k+2} = -\dfrac{a_{k+1} + a_k}{k+2}, \qquad k \geq 0$
- Using it with $a_0 = 1, a_1 = 0$, we obtain solution y_1.
- $a_2 = -\frac{1}{2},\ a_3 = \frac{1}{6},\ a_4 = \frac{1}{12}, \ldots$
- So $y_1(t) = 1 - \frac{1}{2}t^2 + \frac{1}{6}t^3 + \frac{1}{12}t^4 + \cdots$
- Using $a_0 = 0, a_1 = 1$ (solution y_2).
- $a_2 = -\frac{1}{2},\ a_3 = -\frac{1}{6},\ a_4 = \frac{1}{6}, \ldots.$
- So $y_2(t) = t - \frac{1}{2}t^2 - \frac{1}{6}t^3 + \frac{1}{6}t^4 + \cdots.$
- General solution in t: $y = C_1 y_1(t) + C_2 y_2(t)$.
- Substitute back $t = x - 1$ to express the answer in terms of x:

$$y = C_1\left[1-\tfrac{1}{2}(x-1)^2+\tfrac{1}{6}(x-1)^3+\cdots\right]+C_2\left[(x-1)-\tfrac{1}{2}(x-1)^2-\tfrac{1}{6}(x-1)\right.$$

- As $P(x), Q(x)$ have no singularities, this series converges $\forall x$.



Problem 10: Setup ($x_0 = 2$)

- Solve Airy's equation $y'' - xy = 0$ about the ordinary point $x_0 = 2$.
- Here $P(x) = 0$ and $Q(x) = -x$, both entire functions, so every real x_0 is ordinary point.
- Substitute $t = x - 2$, so $x = t + 2$; the equation becomes

$$y'' - (t + 2)y = 0.$$

- Assume $y = \sum_{n=0}^{\infty} a_n t^n$, so $y'' = \sum_{n=0}^{\infty} n(n-1)a_n t^{n-2}$
- Expand $(t + 2)y = ty + 2y = \sum_{n=0}^{\infty} a_n t^{n+1} + 2\sum_{n=0}^{\infty} a_n t^n$
- We must shift all series to a common index k before comparing coefficients.
- Note the first sum ty only starts contributing from $k = 1$ onward.
- This means $k = 0$ must be treated as a separate (initial) case.



Problem 10: Deriving the Recurrence Relation

- Shift y'': let $n = k+2$, giving $\displaystyle\sum_{k=0}^{\infty}(k+2)(k+1)a_{k+2}t^k$.
- Shift ty: let $n+1 = k$, i.e. $n = k-1$, giving $\displaystyle\sum_{k=1}^{\infty}a_{k-1}t^k$ (starts at $k = 1$).
- The term $2y$ is $2\displaystyle\sum_{k=0}^{\infty}a_k t^k$.
- **Case $k = 0$:** only y'' and $2y$ contribute (no t^{-1} term exists)

$$2a_2 - 2a_0 = 0 \implies a_2 = a_0.$$

- **Case $k \geq 1$:** all three series contribute

$$(k+2)(k+1)a_{k+2} - a_{k-1} - 2a_k = 0.$$

- Recurrence relation for $k \geq 1$:

$$a_{k+2} = \frac{a_{k-1} + 2a_k}{(k+2)(k+1)}$$



Problem 10: Final Solution

- Using $a_2 = a_0$ and $a_{k+2} = \dfrac{a_{k-1} + 2a_k}{(k+2)(k+1)}$ with $a_0 = 1, a_1 = 0$ will give solution y_1.
- $a_2 = 1,\ a_3 = \frac{1}{6},\ a_4 = \frac{1}{6},\ a_5 = \frac{1}{15}, \ldots$
- So $y_1(t) = 1 + t^2 + \frac{1}{6}t^3 + \frac{1}{6}t^4 + \cdots$
- Using $a_0 = 0, a_1 = 1$ will give solution y_2.
- $a_2 = 0,\ a_3 = \frac{1}{3},\ a_4 = \frac{1}{12},\ a_5 = \frac{1}{30}, \ldots$
- So $y_2(t) = t + \frac{1}{3}t^3 + \frac{1}{12}t^4 + \cdots$
- General solution in t: $y = C_1 y_1(t) + C_2 y_2(t)$.
- Substitute back $t = x - 2$:

$$y = C_1\left[1+(x-2)^2+\tfrac{1}{6}(x-2)^3+\cdots\right]+C_2\left[(x-2)+\tfrac{1}{3}(x-2)^3+\cdots\right]$$

- Since Airy's equation has no singular points anywhere, this series converges for all x.



Radius of Convergence

- The series solution about x_0 converges at least for $|x - x_0| < R$.
- where R = distance from x_0 to the nearest singular point (real or complex).
- Example: for $(1 - x^2)y'' - 2xy' + n(n+1)y = 0$, singular points are $x = \pm 1$.
- So series about $x_0 = 0$ converges for $|x| < 1$.
- Example: for $y'' - xy = 0$, there are no singular points at all.
- So the series solution converges for every real x.
- This estimate is obtained without solving the equation explicitly.



Series Solution of a Second Order ODE

- We study $y'' + P(x)y' + Q(x)y = 0$ near points where P, Q are not analytic.
- Such points are called singular points of the equation.
- Simple power series $y = \sum a_n x^n$ may fail to exist at a singular point.
- Frobenius (1849–1917) developed a generalized series method for such points.
- The method applies when the singularity is of a mild type, called "regular".
- It extends the power series method by allowing a fractional or negative leading power.
- This is essential for solving Bessel's, Legendre's (at $x = \pm 1$), and Euler equations.
- Frobenius solutions appear widely in physics: heat flow, waves, quantum mechanics.



Recall: Ordinary and Singular Points

- Consider $y'' + P(x)y' + Q(x)y = 0$.
- $x = x_0$ is an **ordinary point** if $P(x)$ and $Q(x)$ are both analytic at x_0.
- At an ordinary point, a power series $y = \sum a_n(x - x_0)^n$ always solves the ODE.
- $x = x_0$ is a **singular point** if $P(x)$ or $Q(x)$ fails to be analytic there.
- Singular points arise where the leading coefficient $a_0(x)$ of the ODE vanishes.
- Not every singular point behaves the same way.
- Some singular points still permit a generalized series solution.
- Others are too severe for any series method to work.
- This leads to the distinction between regular and irregular singular points.
- The distinction is based on the growth rate of $P(x)$ and $Q(x)$ near x_0.



Regular Singular Point: Definition

- Consider $y'' + P(x)y' + Q(x)y = 0$ with a singular point at $x = x_0$.
- Then x_0 is called a **regular singular point** if
- (i) $(x - x_0)P(x)$ is analytic at x_0, and
- (ii) $(x - x_0)^2 Q(x)$ is analytic at x_0
- Equivalently, $P(x)$ has at most a simple pole at x_0 and $Q(x)$ has at most a pole of order two at x_0.
- This mild singularity still permits a Frobenius-type series solution.
- Example: $x^2y'' + xy' + (x^2 - n^2)y = 0$ (Bessel's equation) has $x_0 = 0$ regular.
- Example: Legendre's equation has regular singular points at $x = \pm 1$.



Irregular Singular Point: Definition

- If x_0 is a singular point but not regular, it is called an **irregular singular point**.
- That is, either $(x - x_0)P(x)$ or $(x - x_0)^2 Q(x)$ fails to be analytic at x_0.
- Example: $x^3 y'' + y' + xy = 0$ has $P(x) = 1/x^3$..
- Here $(x - 0)P(x) = 1/x^2$, which is not analytic at $x = 0$.
- Hence $x = 0$ is an irregular singular point of this equation.
- Near an irregular singular point, the Frobenius method generally fails.
- Solutions may involve essential singularities or asymptotic (non-convergent) series.
- Such equations require more advanced techniques beyond this course.
- Fortunately, most classical ODEs of physics have only regular singular points.
- This makes the Frobenius method broadly applicable.



Standard Form Near a Regular Singular Point

- Let $x = 0$ be a regular singular point of $y'' + P(x)y' + Q(x)y = 0$ (shift x_0 to origin if needed).
- By definition, $xP(x)$ and $x^2Q(x)$ are analytic at $x = 0$.
- So we can write $xP(x) = p_0 + p_1x + p_2x^2 + \cdots$ and $x^2Q(x) = q_0 + q_1x + q_2x^2 + \cdots$
- Multiplying the ODE by x^2 gives the convenient form

$$x^2y'' + x[xP(x)]y' + [x^2Q(x)]y = 0.$$

- This form makes the regular singular structure explicit.
- The equation resembles an Euler-Cauchy equation with variable coefficients.
- This motivates seeking solutions of Euler-type times a power series.
- Frobenius proposed exactly such a trial solution.



Frobenius Theorem

- **Theorem (Frobenius):** Let $x = 0$ be a regular singular point of $y'' + P(x)y' + Q(x)y = 0$. Then there exists at least one solution of the form

$$y = x^r \sum_{n=0}^{\infty} a_n x^n, \qquad a_0 \neq 0.$$

- Here r is a constant (real or complex) to be determined.
- The series converges in some interval $0 < x < R$.
- where R is the distance to the nearest other singular point.
- The exponent r need not be a non-negative integer.
- A second linearly independent solution may or may not have the same simple form.
- Its form depends on the relationship between the two roots of the indicial equation (Three different cases will arise !).
- We now derive the equation that determines r.

